

HAWASSA UNIVERSITY



INSTITUTE OF TECHNOLOGY

FACULTY OF INFORMATICS

DEPARTMENT OF INFORMATION SYSTEMS

HUMAN AND COMPUTER INTERACTION PROJECT

On PROJECT TITLE: WEB-BASED FINAL YEAR (INDUSTRIAL) PROJECT
MANAGEMENT SYSTEM FOR HAWASSA FACULTY OF INFORMATICS

GROUP MEMBERS

Name	Id no
Abdi Birhanu	0020/14
Ruth Melaku	1219/15
Meron Endale	0979/15
Natanim Gosaye	1105/15

Submitted to: Dr. Degif T(Phd)

Submission date: 12/21/1015

Project instruction :

- Build prototype for at least two cycles or iterations for a selected project that you are working with.
- The following are the basic components required for the project:
 - **Develop personas.** You can have two or three personas based on the type of application you are working with. Try to look at the basic attributes for a persona and prepare a required level of detail in a poster format.
 - **Develop scenarios** for the application.
 - **Develop prototype.** Show the navigation from the prototype as a user is taken from one part of the screen to the other.
 - **Show the evaluation** of the prototype against the requirements from the persona and scenarios.

1. Develop Personas (The “Who”)

Personas represent the real users of the system. Three personas were developed to reflect the actual stakeholders involved in final-year projects at **Hawassa University**. Each persona focuses on their role, frustrations with the current manual or Telegram-based system, and specific system needs.

Persona 1: Student Project Group (Collaborative Communication)

- **Group Identity:** IS-2025-G04
- **Department:** Information Systems/computer Science/Information technology
- **Faculty:** Informatics
- **Group Leader:** Abebe Tadesse
- **Members:** Sara Kebede, Dawit Tesfaye, Hana Mekonnen
- **Role:** Final Year Student Project Group
- **Technical Skill Level:** Medium

Goals & Responsibilities

- Collaborate effectively as a group and communicate clearly with the advisor.
- Submit project titles and upload official progress reports on time.
- View advisor feedback and contribute to assigned tasks.

Frustrations (Current System)

- Messages are scattered across Telegram threads.
- No central record of advisor feedback or revision history.
- General confusion regarding official submission status.

System Needs

- **Group-based Communication:** A dedicated channel for the group.
- **Functional Separation:** Clear distinction between communication and document submissions.
- **Tracking:** Progress monitoring tied directly to the **Group ID**.

Persona 2: Advisor

- **Name:** Dr. Mesfin Bekele
- **Department:** Information Systems
- **Role:** Project Advisor
- **Technical Skill Level:** High

Goals

- Guide Information Systems student groups efficiently.
- Communicate with all group members simultaneously.
- Track the specific progress of multiple groups using Group IDs.

Frustrations (Current System)

- Receiving messages from multiple groups without a structured format.
- Difficulty identifying which student belongs to which Group ID in a chat list.

System Needs

- **Unified Communication:** Messages organized per Group ID rather than per individual student.
- **Advisor Dashboard:** A screen listing all Information Systems groups under their supervision.
- **Status Indicators:** Visual cues showing which groups are on track or behind.

Persona 3: Department Head

- **Name:** Mr. Dawit Alemu
- **Department:** Information Systems/Computer science/Information technology
- **Role:** Department Head
- **Technical Skill Level:** Medium

Goals

- Monitor project progress across the entire Information Systems department.
- Quickly identify groups that are inactive or delayed.

Frustrations (Current System)

- No centralized, group-level overview of department status.
- Lack of real-time data on project approvals.

System Needs

- **Departmental Dashboard:** High-level overview organized by Group ID.
- **Reporting Tools:** Automated progress and approval reports for all IS projects.



Group Persona



Active Group

GRP-2024-05

Software Engineering
Faculty of Informatics

TECHNICAL SKILL LEVEL

High

Team Composition

3 Members



Abebe Bikila

Group Leader • Full Stack



Sara Tadesse

UI/UX Designer



Kebede Almaz

Backend Developer



GOALS

- Submit functional prototype by June.
- Achieve 'A' grade in final defense.
- Publish a research paper.



DUTIES

- Weekly progress reports.
- Code documentation.
- Advisor meetings.



FRUSTRATIONS

Slow feedback loops

Paperwork overload

Scheduling conflicts



SYSTEM NEEDS

- ✓ Digital submission for all documents.
- ✓ Real-time chat with advisors.
- ✓ Automated meeting scheduling.



Contact Group



Dr. Abebe Kebede

Senior Project Advisor
Hawassa University

 FACULTY OF TECHNOLOGY

 Computer Science Dept

Technical Proficiency

Advanced

Project Management Tools	85%
Mobile App Usage	92%
Digital Collaboration	70%

Primary Goals

-  Efficiently track progress of 5+ student groups simultaneously.
-  Minimize administrative paperwork and focus on mentoring.
-  Ensure timely feedback delivery to students before deadlines.

Frustrations

-  Difficulty tracking email threads for multiple project submissions.
-  Lack of clear visibility on which groups are falling behind schedule.
-  Scattered documentation across physical papers and various digital folders.

System Needs


Centralized
Dashboard


Real-time Alerts


PDF Reports


In-app Chat

← Department Head Persona



Mr. Dawit Alemu

Head of Computer Science Dept.

FACULTY
IoT Institute

EXPERIENCE
12 Years

 **Technical Proficiency** Intermediate

3/5

Comfortable with basic digital tools and email, but prefers intuitive dashboards over complex systems.

Primary Goals

- ✓ Efficiently track student project progress without manual follow-ups.
- ✓ Reduce paperwork and emails for advisor assignments.
- ✓ Ensure timely grade submissions and academic compliance.

Key Frustrations

- ❗ Lost project files buried in endless email threads.
- ❗ Difficulty monitoring advisor workloads leading to imbalance.
- ❗ Lack of centralized reporting for departmental meetings.

System Needs

-  Dashboard for real-time status overview
-  Automated deadline notifications
-  Digital signature integration



HAWASSA UNIVERSITY

2. Develop Scenarios (The “What”)

Scenarios describe the specific steps and context in which the users interact with the system. These narratives demonstrate how the prototype solves the problems identified in the persona phase.

Scenario 1: Collaborative Student Communication and Submission

- **User:** Student Project Group (IS-2025-G04)
- **Action:** The group members log into the system using their shared project context.
- ✓ **Abebe Tadesse** (Leader) navigates to the "Submissions" tab and uploads the bi-weekly progress report. Simultaneously, **Sara Kebede** uses the "Group Communication Panel" to send a technical clarification question to their advisor.
- **Outcome:** The advisor's response is archived and remains visible to all members of the group, ensuring no one misses the feedback.

Scenario 2: Advisor Interaction with Group

- **User:** Dr. Mesfin Bekele (Advisor)
- **Action:** Dr. Mesfin logs into the **Advisor Dashboard**. He sees a list of all Information Systems groups he is supervising, organized by **Group ID**.
- ✓ He selects **IS-2025-G04** to view their specific activity.
- ✓ He reviews the recently uploaded progress report, types a response to Sara's question, and clicks the "Approve" button for the current milestone.
- ✓ **Outcome:** The group is instantly notified of the approval, and their status updates from "Pending" to "Approved."

Scenario 3: Department Head Oversight

- **User:** Mr. Dawit Alemu (Department Head)
- **Action:** Mr. Dawit logs into the **Department Head Dashboard**.
- ✓ He filters the view by the **Information Systems** department to monitor all active project groups.
- ✓ He reviews the overall approval status and progress timelines for each **Group ID** to identify any groups that have not submitted reports on time.
- ✓ **Outcome:** He ensures academic compliance across the department and can proactively contact advisors for groups that are falling behind.

3. Build the Prototype in Two Cycles (The “How”)

The development process follows an iterative approach, moving from structural navigation in the first cycle to refined usability and visual indicators in the second.

Cycle 1: Low-Fidelity Prototype (Navigation Focus)

The primary goal of this cycle is to establish the user flow and ensure all stakeholders can access their required features.

Core Screens Included:

- **Login Page:** Featuring the Hawassa University logo.
- **Student Group Dashboard:** Displays Group ID, Department (Information Systems), Leader badge, and Member list.
- **Group Communication Page:** A shared portal where all members can message the advisor.
- **Submit Project Title Page:** A restricted-access page for the Group Leader only.
- **Upload Group Progress Page:** A restricted-access page for the Group Leader only.
- **Advisor Dashboard:** A summary view listing all Information Systems groups by Group ID.
- **Review Group Progress & Communication Page:** The advisor’s interface for responding and grading.
- **Department Head Dashboard:** A high-level Information Systems overview.

Navigation Examples:

- **Access:** Login → Role-based Dashboard (Student, Advisor, or Head).
- **Collaboration:** Student Dashboard → Communication → Advisor Response.
- **Workflow:** Student Dashboard → Upload Progress → Advisor Review.

Cycle 2: Medium-Fidelity Prototype (Usability Improvement)

The second iteration focuses on enhancing the user experience (UX) by adding visual cues, status tracking, and consistent branding.

Key Enhancements:

- **Role Identification:** Added a distinct **Visual Group Leader badge** to clarify submission permissions.
- **Enhanced Notifications:** All alerts are now clearly labeled with the specific **Group ID and Department** to prevent advisor confusion.
- **Progress Tracking:** Implementation of a **Progress Timeline** for "Project I" and "Project II" specific to each Information Systems group.

- **Status Indicators:** Added color-coded labels for immediate recognition of task status:
 - ✓ **Pending** (Yellow)
 - ✓ **Approved** (Green)
 - ✓ **Rejected** (Red)
 - ✓ **Overdue** (Dark Red)
 - ✓ **Interface Consistency:** Standardized the layout across all dashboards to reduce the user's cognitive load.

Cycle 1 prototype

Login page

Student group dashboard

Group communication page

The screenshot shows a web interface for a group workspace. On the left is a sidebar with navigation links: Dashboard, Submissions, Calendar, and Communication (highlighted). The main area is titled 'Group 42 Workspace' and shows a chat history with Dr. Abebe and Sara Kebede. The chat messages include a welcome message, a request for a proposal, a discussion about methodology, and a confirmation of updates. A pinned notice on the right states: 'Important: Review meeting scheduled for Friday at 2 PM. Please submit draft beforehand.' Below this, project info shows 'Group 42' with 4 members and 'Industrial Project' with course code CS401. Quick actions include 'Submit Progress Report 1' and 'Schedule Meeting'. Recent files list 'Project_Proposal_v2.pdf' and 'Requirements_Spec.docx'. At the bottom, there is a text input field for messages and a 'Settings' link.

Submit project title page

The screenshot shows a web form for submitting a project title. The header includes 'Hawassa University Project Management System' and 'Abebe Tadesse Group Leader'. The breadcrumb trail is 'Dashboard / Project Submission / Submit Title'. The main heading is 'Submit Project Title' with a subtitle 'Final Industrial Project - Group Leader Action'. An 'Important Submission Guidelines' section states: 'Please ensure your title is approved by your potential advisor before submission. Once submitted, changes may require administrative approval from the department head. Your title must be unique and descriptive (max 150 characters)'. The form contains three input fields: 'Proposed Project Title' (with a placeholder 'e.g., Automated Inventory Management System for Hawassa Textile Factory' and a character count '0 / 150 characters'), 'Group ID' (with value 'HU-CS-2023-G04'), and 'Department' (with value 'Computer Science'). At the bottom are 'Cancel' and 'Submit Title' buttons. A footer note says 'Need help? Contact the Department Coordinator.'

Upload group progress page

Hawassa University PMS

DashboardProjectsGradesResourcesAbebe TadesseGroup Leader

Home / Project / Progress Report

Upload Group ProgressWEEK 12

Final Industrial Project - Group A-24

Logged in as Group Leader

Submission Guidelines

- Upload the report in PDF or DOCX format only.
- Maximum file size is 25MB.
- Ensure the file follows the naming convention: GroupID_WeekNumber_Progress.pdf

Click to upload or drag and drop

PDF, DOCX (max. 25MB)

Browse Files

Group_A24_Week12_Report.pdf

2.4 MB · Ready to submit

Advisor Comments / Notes (Optional)

Enter any specific details regarding this week's progress...

CancelSubmit Report

Current Status

Draft Created

Oct 24, 10:00 AM

Ready to Upload

Action Required

Submitted

Pending

Graded

→ / 100

Assigned Advisor

Dr. Mesfin Bekele

Industrial Engineering Dept.

Contact Advisor

Submission History

Date	File Name	Version	Status	Action
Oct 24, 2023	Week_11_Report.pdf	v1.0	Graded	VIEW FEEDBACK
Oct 17, 2023	Week_10_Report.docx	v2.1	Graded	VIEW FEEDBACK

Advisor dashboard page

Project Management

Hawassa University

Dashboard

Defense Schedule

Past Projects

Student Messages1

Settings

Log Out

Advisor Dashboard

Welcome back, Dr. Mesfin Bekele. You have 3 pending project reviews.

Broadcast to Students

Assigned Groups12

Pending Reviews3

Next Defense DateOct 24

Search by Group ID, Title, or Student Name

All GroupsOn TrackNeeds ReviewBehind S

GROUP ID	PROJECT TITLE	TEAM MEMBERS	PHASE	STATUS	ACTION
IS-2024-05	AI-Driven Crop Disease Detection Updated 2h ago		Implementation	On Track	
IS-2024-12	Blockchain for Land Registry Submitted yesterday	+1	Proposal	Needs Review	Review
IS-2024-08	E-Learning Platform for Rural Areas No updates in 5 days		Literature Review	Behind Schedule	
IS-2024-02	Smart Traffic Management System Updated 1 day ago		Implementation	On Track	
IS-2024-15	Hospital Management System v2 Submitted 3h ago		Requirement Analysis	Needs Review	Review

Showing 1-5 of 12 groups

Review progress page



Hawassa University PMS
ADVISOR PORTAL

Dashboard

My Groups

Schedule

Grading



Dr. Mesfin Bekele
Advisor

Home / My Groups / Group 4 / Review

Smart Inventory System (Group 4)

Pending Review

Submitted Oct 12, 2023

View Members

Current Submission: Milestone 3

Download All



System_Design_Doc_v2.pdf

2.4 MB

DESCRIPTION

This document covers the architectural design, database schema (ERD), and interface mockups for the Smart Inventory System. We have addressed the feedback from the previous milestone regarding the normalization of the "Products" table.

Submitted By

Abebe Kebede

Version

Revision 2.0

Review Feedback

Enter your official feedback for the record here...

Approve Milestone

Request Revision

Progress Timeline

✓

Project Proposal
Approved on Sep 15, 2023

✓

Requirements Analysis
Approved on Sep 30, 2023

📄

System Design Document
Pending Advisor Review

<>

Implementation Phase 1
Locked

Messages

Q&A Board

TODAY



Abebe Kebede
10:42 AM

Good morning sir, regarding the ER diagram, should we include the audit logs table in the main schema or keep it separate?

10:45 AM

Dr. Mesfin Bekele

Keep It separate for now to avoid cluttering the main transactional flow. I've added a note in the document as well.



Sara Tadesse
11:02 AM

Noted: We will update the submission.

Type your message...

Press Enter to send, Shift+Enter for new line

GROUP MEMBERS

2 Online



Department head dashboard


Hawassa University PMS

Dashboard
Projects
Advisors
Settings


Department Head Dashboard

Mr. Dawit Alemu • Information Systems Department

Academic Year: 2023/2024

Total IS Groups
24

Pending Approvals
5
Needs attention

Projects At Risk
3
Behind schedule

Completed
12

All Status
Proposal Phase
Implementation

GROUP ID	PROJECT TITLE	ADVISOR	PROGRESS	STATUS	ACTION
IS-24	Smart Inventory System Mobile & Web App	Dr. Alemu	65%	Implementation	
IS-25	E-Learning Platform Web Application	Ms. Tigist	15%	Proposal Review	Review
IS-28	Hospital Mgmt System Legacy System Upgrade	Mr. Kebede	40%	At Risk	
IS-30	Attendance via Facial Rec. AI / ML Project	Ms. Aster	100%	Completed	
IS-32	Campus Navigation Mobile App	Dr. Alemu	80%	Testing	

Showing 1 to 5 of 24 entries

Deadlines

- Oct 15 - Completed
Proposal Submission
- Dec 20 - Upcoming
Chapter 1-3 Review
- Feb 10 - Future
Final Defense

Activity

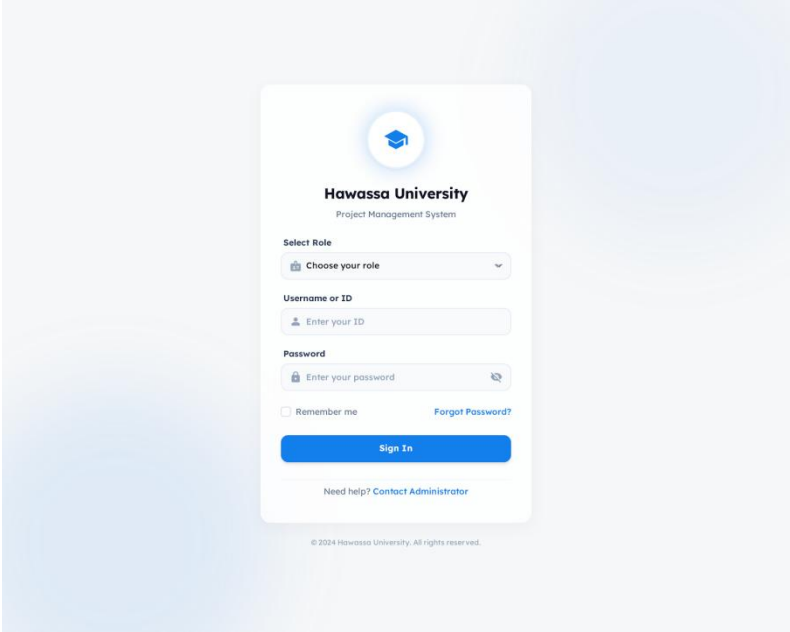
- Group IS-25 submitted
Proposal
2 hours ago
- Advisor Approved IS-32
Chapter 2
5 hours ago
- IS-28 missed progress
report
Yesterday

QUICK ACTIONS

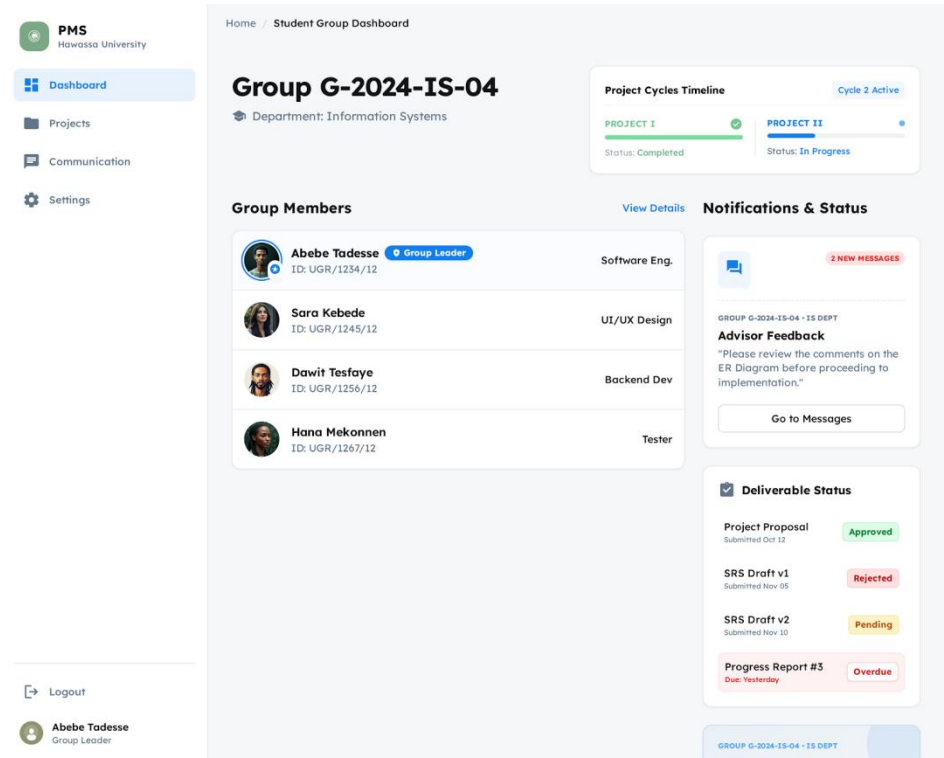
Broadcast Message

+ New Group

Second cylcle prototype



Student group dashboard



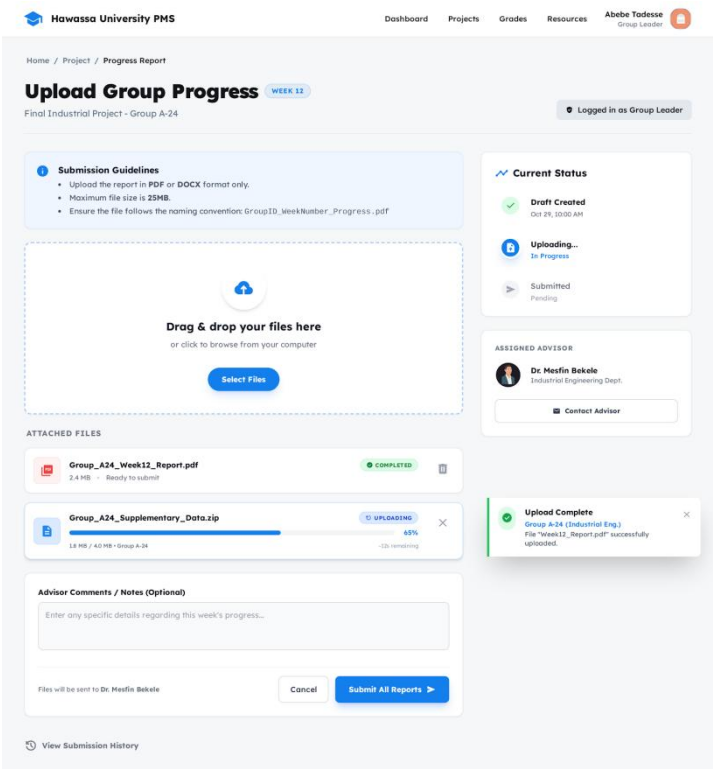
Group communication page

The screenshot shows a web interface for a group workspace. On the left is a sidebar with navigation links: Dashboard, Submissions, Calendar, and Communication (highlighted). The main area is titled 'Group 42 Workspace' and shows a chat conversation. The chat starts with a message from Dr. Mesfin Bekele (Advisor) welcoming the group and asking for an initial proposal by Friday. Sara Kebede responds with a thank you and a question about the methodology section. Dr. Mesfin Bekele replies that the methodology should be in the appendix. Sara Kebede then updates the draft. A yellow warning banner at the bottom of the chat area states: 'Please do not upload full project documents here. Use the Submissions tab for official deliverables.' On the right side, there is a 'Pinned Notice' about an upcoming deadline, a 'Context' section showing group details, 'Quick Actions' for submitting progress reports and scheduling meetings, and a 'Recent Shared Files' section listing two documents: 'Project_Proposal_v2.pdf' and 'Requirements_Spec.docx'.

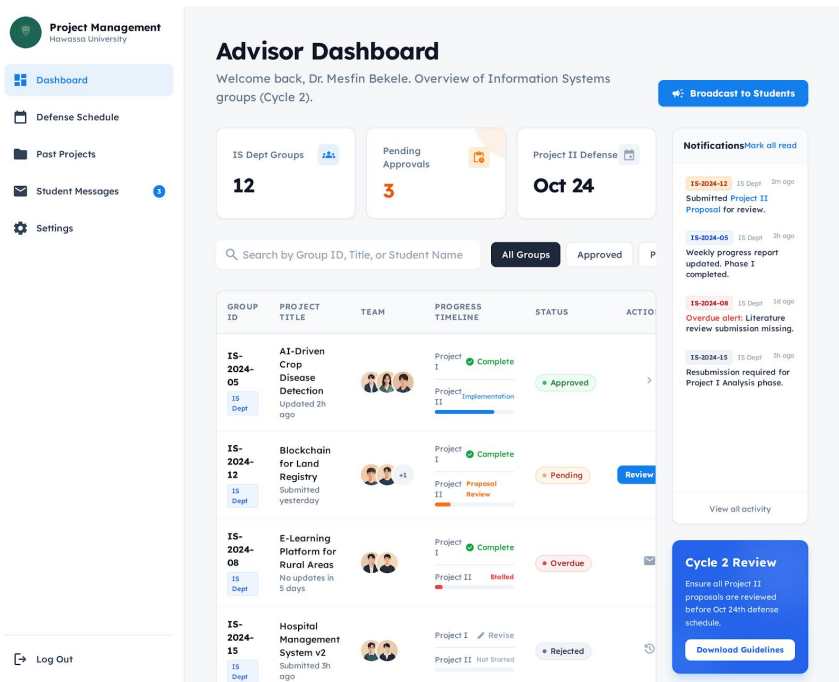
Submit project title page

The screenshot shows a web interface for submitting a project title. At the top, there is a progress bar with four steps: Team Setup (completed), Submit Title (current step), Advisor, and Approval. The main heading is 'Submit Project Title' for the 'Final Industrial Project - Cycle 2'. Below this, there are 'Submission Guidelines' stating that the title must be approved by the advisor and be unique and descriptive (max 150 characters). A 'Proposed Project Title' input field contains the text 'e.g., Automated Inventory Management System for Hawassa Textile Factory'. Below the input field, there are fields for 'Group ID' (HU-CS-2023-G04) and 'Department' (Computer Science). At the bottom, there are 'Cancel' and 'Submit Title' buttons. A footer note says: 'Need help? Contact the Department Coordinator.'

Upload group progress page



Advisor dashboard page



Review progress page

Project Management

Hawassa University

Dashboard

Defense Schedule

Past Projects

Student Messages

Settings

Log Out

Adviser Dashboard

Welcome back, Dr. Mesfin Bekele. Overview of Information Systems groups (Cycle 2).

IS Dept Groups

12

Pending Approvals

3

Project II Defense

Oct 24

Broadcast to Students

Search by Group ID, Title, or Student Name

All Groups

Approved

Pending

GROUP ID	PROJECT TITLE	TEAM	PROGRESS TIMELINE	STATUS	ACTION
IS-2024-05	AI-Driven Crop Disease Detection		Project I Complete Project II Implementation	Approved	
IS-2024-12	Blockchain for Land Registry		Project I Complete Project II Proposal Review	Pending	Review
IS-2024-08	E-Learning Platform for Rural Areas		Project I Complete Project II Stalled	Overdue	
IS-2024-15	Hospital Management System v2		Project I Revise Project II Not Started	Rejected	

Notifications

Mark all read

IS-2024-12

IS Dept

2m ago

Submitted Project II Proposal for review.

IS-2024-05

IS Dept

2h ago

Weekly progress report updated. Phase I completed.

IS-2024-08

IS Dept

1d ago

Overdue alert: Literature review submission missing.

IS-2024-15

IS Dept

3h ago

Resubmission required for Project I Analysis phase.

View all activity

Cycle 2 Review

Ensure all Project II proposals are reviewed before Oct 24th defense schedule.

Download Guidelines

Department head dashboard

Hawassa University PMS

Dashboard

Projects

Advisors

Settings

Department Head Dashboard

Mr. Dawit Alemu • Information Systems Department

Academic Year: 2023/2024

Total IS Groups

24

Pending Approval

5

Requires review

Overdue Projects

3

Behind schedule

Approved & Active

16

Search Group ID or Title

All Status

Pending

Approved

Overdue

GROUP ID	PROJECT TITLE	ADVISOR	TIMELINE	STATUS	ACTION
IS-24	Smart Inventory System	Dr. Alemu	65%	Approved	
IS-25	E-Learning Platform	Ms. Tigist	15%	Pending	Review
IS-28	Hospital Mgmt System	Mr. Kebede	40%	Overdue	
IS-30	Attendance via Facial Rec	Ms. Aster	100%	Approved	
IS-32	Campus Navigation	Dr. Alemu	5%	Rejected	

Showing 1 to 5 of 24 entries

Deadlines

Oct 15 - Completed

Proposal Submission

Dec 20 - Upcoming

Chapter 1-3 Review

Feb 10 - Future

Final Defense

Notifications

Mark all read

IS-25

IS DEPT

2 hours ago

Proposal submitted for review

IS-30

IS DEPT

5 hours ago

Advisor approved Chapter 2

IS-28

IS DEPT

Yesterday

Progress report overdue (2 days)

IS-32

IS DEPT

2 days ago

Project topic rejected by committee

QUICK ACTIONS

Broadcast Message

New Group

4. Prototype Evaluation (The “Why”)

This section evaluates the final design against the original user requirements identified in the persona and scenario phases. It demonstrates how the system successfully addresses the needs of the stakeholders at Hawassa University.

Requirement vs. Evaluation

Requirement 1: Advisor Management Efficiency

- **User Need:** The advisor needs to manage multiple Information Systems groups simultaneously without losing track of individual group progress.
- **Evaluation:** Through the Advisor Dashboard, groups are **clearly labeled by Group ID and Department**. This allows for immediate identification, enabling the advisor to filter by group and supervise projects efficiently.

Requirement 2: Group Accountability & Collaboration

- **User Need:** Students require a space for shared communication but need clear ownership over official project submissions to avoid errors.
- **Evaluation:** The prototype features a dual-access system. While **all group members** can participate in the communication panel with the advisor, the "Submit" and "Upload" functions are restricted to the **Group Leader**, ensuring clear accountability for official milestones.

Refinement: Improvement from Cycle 1 to Cycle 2

The transition from a low-fidelity to a medium-fidelity prototype focused on bridging the gap between basic functionality and a seamless user experience:

- **Clarity & Coordination:** Iteration 2 introduced **automated notifications** and **status indicators** (Pending, Approved, Rejected). These are directly linked to the **Group ID**, preventing the "scattered message" issue found in the previous Telegram-based system.
- **Visual Progress:** The addition of **Progress Timelines** for Project I and Project II provides a road map for students and a quick compliance check for the Department Head, ensuring all Information Systems projects remain on schedule.